

VISHWASWARUP RATH

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EDUCATION

Manipal University Jaipur

October 2024 – June 2028

• B.Tech in Computer Science and Engineering with Specialization in AI/ML

CGPA: 8.95

EXPERIENCE

Trainee | Centre for Artificial Intelligence and Robotics (CAIR), DRDO, Bengaluru

June 2026 – July 2026

- Engineered a RAG-based system to process and retrieve information across **5+ multi-modal data types** (reports, documents, news articles, images), streamlining intelligence analysis workflows.
- Designed **3 core pipeline components** – information extraction, semantic retrieval, and structured knowledge generation – to support end-to-end intelligence analysis.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, TypeScript

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Ollama, NLP, Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search, OpenCV, YOLOv8

Data Science: Pandas, NumPy, Matplotlib

Backend: FastAPI, Flask, Node.js, REST APIs, Async Processing

Frontend: React, Next.js, Tailwind CSS

Databases: PostgreSQL, MongoDB, SQLite, Supabase, ChromaDB (Vector Database)

Data Engineering: ETL Pipelines, Web Scraping, Scrapy, BeautifulSoup, Selenium

Cloud & DevOps: AWS, Docker, Git, GitHub, CI/CD

Software Engineering: System Design, Microservices, Scalable Architecture

FEATURED PROJECTS

Artefax – Gen AI Content Transformation Platform [GitHub] [Live Demo]

September 2026

- Architected an AI content-transformation platform that extracts a single structured understanding of a source document once, then generates **4 distinct output types** - Executive Summary, LinkedIn Post, Advisory, and Presentation, from that shared understanding to guarantee factual consistency across outputs.
- Designed a **Content Brief** and **Transformer Registry** pattern so every AI-generated artifact is validated against strict **Pydantic schemas** before a separate, deterministic rendering layer converts it into real downloadable files (**PDF, PPTX, TXT**) via ReportLab and python-pptx.
- Built with **Next.js/TypeScript** frontend and **Python/FastAPI** backend, using the **Gemini API** for structured, schema-validated generation and SQLite for metadata persistence; deployed both services on Render.

IntelliRAG – Incident-Centric RAG for Terrorism Analysis [GitHub]

June 2026

- Building a multi-modal RAG platform ingesting **5 document/media types** (PDFs, news articles, scanned documents, images, reports) into structured, searchable intelligence records.
- Architected a **6-stage pipeline** – OCR, information extraction, entity and relationship mapping, semantic retrieval, layered summarization, and analyst-oriented querying – for intelligence analysis.
- Leveraging Python, local LLMs, vector search, SQLite, and scalable data-processing workflows to support threat assessment and intelligence-driven decision making.

Stock Recommender System [GitHub]

August 2026

- Built a full-stack personalized stock recommendation platform on a **three-tier architecture** (React frontend, FastAPI REST backend, PostgreSQL database) with modular authentication, recommendation, admin, and health-check services.
- Implemented **JWT-based authentication (HS256)** with bcrypt password hashing, dependency-injected database sessions, and **SQLAlchemy 2.0 ORM** with **Alembic**-managed schema migrations across Users, Stocks, and Recommendations entities.
- Developed the frontend in **React 18 + Vite** with centralized JWT-aware API utilities and Context-based auth state, serving personalized recommendations with confidence scores through a protected `/recommendations` endpoint.

Biterate – Restaurant Review Sentiment Analyzer [GitHub] [Live Demo]

April 2026

- Engineered an end-to-end NLP pipeline for restaurant review sentiment analysis featuring **emoji-aware preprocessing**, tokenization, lemmatization, stopword removal, negation handling, and **TF-IDF vectorization with unigram, bigram, and trigram features**.

- Trained, tuned, and benchmarked **5 models** — Logistic Regression (GridSearchCV), Naive Bayes, Linear SVC, XGBoost, and a **Soft Voting Ensemble** — using **5-fold Stratified Cross Validation**, achieving up to **88% accuracy** and an **0.87 F1-score**. Deployed the best-performing model through an interactive **Flask**-based dashboard for real-time sentiment prediction and review analytics.

Vision AI – Voice-Controlled Personal Assistant [GitHub]

October 2025

- Built a Python-based voice assistant from scratch integrating **SpeechRecognition** for live speech-to-text and the **Google Gemini API** for natural-language understanding and conversational responses, with macOS **TTS** for natural spoken replies.
- Implemented voice-controlled browsing and media commands (e.g. web search, YouTube, Spotify playback) via the `webbrowser` module, and secured API credentials using **dotenv**.
- Added contextual memory of prior actions so the assistant can respond to follow-up commands naturally, similar to a real personal assistant.

Clubs and Co-Curricular

First Year Co-ordinator | MUJ ACM SIGAI Student Chapter

Nov 2024 – May 2025

- Selected among 14 candidates from 600+ applicants to coordinate college-level technical competitions and workshops, while contributing to website management and hosting initiatives.

Core Technical Team | AWS Cloud Club – MUJ

Nov 2025 – May 2026

- Built and maintained the official AWS Cloud Club MUJ website while contributing to the planning and execution of technical workshops, speaker sessions, and community events.

ACHIEVEMENTS AND CERTIFICATIONS

- Awarded the **Dean's List** for outstanding academic performance by securing the highest GPA across **two consecutive semesters** (1st and 2nd).
- Selected for **Amazon ML Summer School 2026**, a highly competitive machine learning program, among **3,000 participants** from over **134,000 applicants**.
- Successfully completed the **Machine Learning Specialization** by **Stanford University** and **DeepLearning.AI**, covering supervised learning, advanced learning algorithms, and unsupervised learning.
- Successfully completed the **Data Structures and Algorithm Design** course offered by **NPTEL**, earning the **ELITE** certification.
- Secured **2nd Place** at the **Innoverse Hackathon** by developing an innovative solution in a competitive team environment.